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OMM Lab #42 Pre-Lab Lecture: Popliteal Fossa Release, Thoracic Pumps, Rib FPR, Posterior Thoracic Counterstrain, Cardiopulmonary Osteopathic Cases

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Session Objectives

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1. Perform new techniques: Popliteal Fossa Release, Thoracic Pumps, Rib FPR, and Posterior Thoracic Counterstrain
2. Understand indications and contraindications to treatment of cardiopulmonary somatic dysfunctions in the acute vs. chronic setting
3. Explain the reasoning for various treatment goals for the cardiopulmonary patient, as described in the required readings.



Popliteal Fossa Release, Supine

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Indications and Physiologic Goal: This technique is indicated to improve lymphatic and venous drainage from the lower extremities (knee, calf, ankle, and foot) and to release any fascial restriction(s) of the popliteal fossa.
Ex: Left Popliteal Congestion

Technique:

1. The patient lies supine with legs extended on table.
2. The physician, facing the head of the table, sits at the side to be treated.
3. The physician's medial hand reaches around to the medial aspect of the popliteal fossa as the lateral hand grasps the lateral aspect of the popliteal fossa.
4. The physician palpates for any fascial restrictions, including cephalad, caudad, medial, and lateral.
5. The physician engages the tissues with an anterior force through the fingertips while engaging any fascial barriers (e.g., cephalad, caudad, medial, lateral) until resistance is met.
6. The physician can attempt to feel a fluid ebb and flow with a resultant release or add a release- enhancing mechanism by instructing the patient to inhale and exhale deeply. Force is directed into the barrier until no further excursion or relaxation of the tissues is possible.
7. This technique is repeated on the opposite side as needed.



Step 1-3: Setup and Hand Placement



Step 4: Determining Barriers



Step 5: Direct Myofascial Release

Thoracic (Lymphatic) Pump

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Indications: This technique is indicated for infection, fever, lymphatic congestion, rales, and chronic productive cough; also preventive, it may increase titers postvaccination

Physiologic Goal: The goal is to accentuate negative intrathoracic pressure, increase lymphatic return, loosen mucus plugs via the vibratory component, and potentially stimulate the autoimmune system.

Technique:

1. The patient lies supine with the head turned to one side (to avoid breathing or coughing into the face of the physician) and with the hips and knees flexed and the feet flat on the table.
2. The physician stands at the head of the table with one foot in front of the other.
3. The physician places the thenar eminences inferior to the patient's clavicles with the fingers spreading out over the upper rib cage. For female patients, the physician places the hands more midline over the sternum. (*Another approach is to have patients place their hands over their chest, and the physician places their hands on top of the patients hands)*
4. The patient is instructed to take a deep breath and exhale fully.
5. During exhalation, the physician increases the pressure on the anterior rib cage, exaggerating the exhalation motion.
6. At end exhalation, the physician imparts a vibratory motion to the rib cage at two compressions per second (pulsed arrows)
7. Should the patient need to breathe, pressure is relaxed slightly, but the compressions are continued for several minutes.

Thoracic (Lymphatic) Pump continued



Hand Position



Modified Hand Position



Two compressions per second

Thoracic (Lymphatic) Pump Exaggerated Respiration Method

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Indications: This technique is indicated for infection, fever, lymphatic congestion, rales, and chronic productive cough; it is also preventive.

Physiologic Goal: The goal is to accentuate negative intrathoracic pressure and increase lymphatic return.

Technique:

1. The patient lies supine with the head turned to one side (to avoid breathing or coughing into the face of the physician) and with the hips and knees flexed and the feet flat on the table.
2. The physician stands at the head of the table with one foot in front of the other.
3. The physician places the thenar eminences inferior to the patient's clavicles with the fingers spreading out over the upper rib cage. For female patients, the physician places the hands more midline over the sternum .
4. The patient is instructed to take a deep breath and exhale fully.
5. During exhalation, the physician increases the pressure on the anterior rib cage, exaggerating the exhalation motion (arrow).
6. During the next inhalation, the physician releases the pressure (upward arrow) and then reinstates it (downward arrow) with the next exhalation.
7. This version of the thoracic pump may be repeated for 5 to 10 respiratory cycles. This may hyperventilate the patient, and light-headedness and dizziness are fairly common.

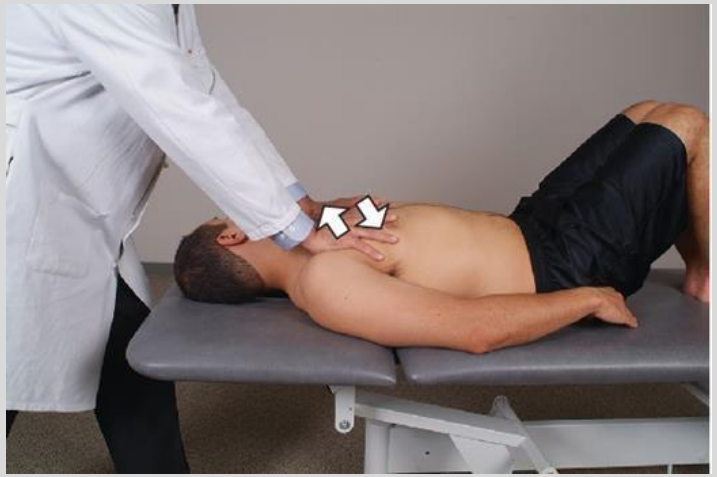
Thoracic (Lymphatic) Pump Exaggerated Respiration Method Continued



Hand Position/ Modified Hand Position



Pressure with exhalation



Release pressure on inhalation and reinstate it on exhalation

Thoracic (Lymphatic) Pump Atelectasis Modification Method

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Indications: This technique is indicated for atelectasis.

Physiologic Goal: The goal is to accentuate the negative phase of respiration and clear mucus plugs.

Technique:

1. The patient lies supine with the head turned to one side (to avoid breathing or coughing into the face of the physician) and with the hips and knees flexed and feet flat on the table.
2. The physician stands at the head of the table with one foot in front of the other.
3. The physician places the thenar eminences inferior to the patient's clavicles with the fingers spreading out over the upper rib cage. For female patients, the physician places the hands more midline over the sternum.
4. The patient is instructed to take a deep breath and exhale fully.
5. During exhalation, the physician increases the pressure on the anterior rib cage, exaggerating the exhalation motion.
6. During the next several inhalations, the physician maintains heavy pressure on the chest wall.
7. On the last instruction to inhale, the physician suddenly releases the pressure, causing the patient to take a very rapid, deep inhalation, inflating any atelectatic segments that may be present.

Thoracic (Lymphatic) Pump Atelectasis Modification Method

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Hand Position/ Modified Hand Position



Exaggerating exhalation, restricting inhalation



Sudden release of pressure

Unilateral Thoracic (Lymphatic) Pump Side Modification Method

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Indications: This technique is indicated for atelectasis.

Physiologic Goal: The goal is to accentuate the negative phase of respiration and clear mucus plugs.

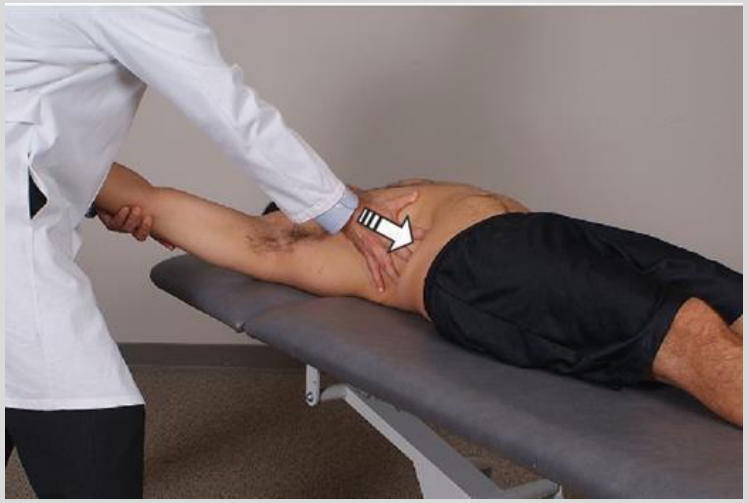
Technique:

1. The patient lies supine, and the physician stands at the side of the table at the level of the patient's rib cage. The patient's hips and knees may be flexed to enhance the effect.
2. The patient's arm is abducted 90 degrees or greater, and the physician exerts traction on the arm with the cephalad hand.
3. The physician places the caudad hand over the lower costal cartilages with the fingers following the intercostal spaces
4. The patient is instructed to take a deep breath and exhale fully.
5. At end of exhalation, a percussive or vibratory motion (arrow) is exerted by the physician at two per second.
6. Should the patient feel the need to breathe, pressure is released just enough to permit easy respiration and the vibratory motion continued.
7. This technique is continued for several minutes. It should be repeated, when possible, on the opposite side of the chest.

Unilateral Thoracic (Lymphatic) Pump Side Modification Method Continued



Steps 1-3: setup and hand placement



Step 5: percussive or vibratory motion

Treatment of the First Rib with Facilitated Positional Release (FPR)

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Fig 73.1

- *Patient position*: supine.
- *Physician position*: standing beside the table, on the side of the dysfunction, facing the head of the table.
- *Technique*:
    1. The physician places their near hand over the patient's rib with the fingers on the posterior aspect of the first rib (right hand on right rib). The hand placement may be modified to treat soft tissue only or at the costovertebral junction to obtain first rib articular release. Their fingers should identify the area of greatest tension to treat the muscle.
    2. The physician bends the patient's elbow and places their far hand over the patient's flexed elbow and then brings the humerus up to 90 degrees of flexion.
    3. The physician compresses down on the patient's elbow with the force directed toward their monitoring fingers (Fig. 73-1).

Treatment of the First Rib with Facilitated Positional Release (FPR) Continued

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Fig 73.2

4. Maintaining this compression, the physician adds internal rotation of the shoulder joint. This is accomplished by placing the patient's right forearm on the ventral aspect of the physician's near forearm and turning the patient's forearm outward by a caudal motion of the physician's forearm. This should create tissue release or articular motion.

5. The position is held for 3 seconds.

6. Maintaining the compressive torsional force, the physician adducts the patient's arm toward the midline and then circumducts the arm down to the tabletop until the arm is along the patient's side (Fig. 73-2).

7. The position is released, and the dysfunction is reevaluated.

Anterior Rib Cage and Costochondral Dysfunctions or Muscle Hypertonicity



Fig 73.3

Example: Fourth, Left, Costochondral Dysfunction

Patient position: seated.

Physician position: standing behind the patient.

Technique:

1. The physician places their right arm around the front of the patient with the index finger of that hand on the tender tissue or on the rib somatic dysfunction.
2. The physician's left hand is on the patient's cervicothoracic junction, with the left arm resting on the superior aspect of the patient's shoulder and their elbow over the acromion process.
3. The patient is instructed to sit up straight until the thoracic kyphosis flattens slightly.
4. The physician applies compression at the cervicothoracic junction and left shoulder, directly toward the floor (Fig. 73-3). (Do not allow forward bending.)

Anterior Rib Cage and Costochondral Dysfunctions or Muscle Hypertonicity Continued



Fig 73.4

5. Maintaining the compressive force, the physician bends the patient's body forward to the monitoring finger. (The patient may be told to drop their head forward.)

6. The physician adds left side-bending until motion is felt at the monitoring finger. Rotation may have to be added (Fig. 73-4).

7. The position is held for 3 seconds and then released, and the dysfunction is reevaluate

Posterior rib Dysfunctions or Muscle Hypertonicity

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Patient Prone Technique
Patient position: prone.
Physician position: standing at the side of the table opposite the somatic dysfunction.

Technique:
1. The physician reaches their caudad hand across to the patient's contralateral shoulder.
2. They place the index finger of their cephalad hand on the costotransverse articulation to monitor motion.
3. With the hand on the shoulder, they introduce a compressive force toward the patient's feet, introducing side-bending of the contralateral thoracic spine until motion is felt at the somatic dysfunction.
4. A rotational or torsional force may be added until the tissue texture softens.
5. This position is held for 3 to 5 seconds.
6. The shoulder is returned to a neutral position, and the somatic dysfunction is reassessed.

Posterior rib Dysfunctions or Muscle Hypertonicity



Posterior rib Dysfunctions or Muscle Hypertonicity

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Patient Seated Technique

Patient position: seated on the edge of the table or on a stool.

Physician position: standing behind the patient on the side of the somatic dysfunction at a 90 degree angle with the patient's back.

Technique:

1. The physician monitors the somatic dysfunction with the hand farthest from the patient.
2. The patient is asked to sit up straight and push their chest forward.
3. The physician's flexed elbow of their near arm is placed anterior to the patient's shoulder with the forearm on top of the shoulder. With this arm, they compress the thoracic region downward and side-bends the region to the side of the dysfunction. With their elbow, they add a rotary force drawing the patient's shoulder posteriorly localizing to the site of the somatic dysfunction.
4. This position is held for 3 to 5 seconds.
5. The forces are released, and the somatic dysfunction is reassessed.

Note: This technique is very similar to that performed for the posterior somatic dysfunctions of the thoracic spine performed in the seated position. The major difference is the point at which the monitoring finger detects the forces at the site of the dysfunction.

Lateral Rib Dysfunctions or Muscle Hypertonicity

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Example: Fifth Right Rib

Patient position: seated.

Physician position: standing on the patient's right side. Initially, the physician is oriented approximately 90 degrees away from the patient (their back near the patient's right shoulder).

Technique:

1. The physician places their left axilla over the patient's right shoulder as close to the cervicothoracic junction as possible.
2. The physician uses their left hand to grasp the patient's right elbow. The patient's elbow is flexed, and the shoulder is brought to approximately 45 degrees of flexion and abduction.
3. The physician uses their right index finger to monitor the dysfunction.
4. The patient is instructed to sit up straight.
5. The physician adds a compressive force from their axilla down onto the patient's shoulder parallel to the spine.
6. A slight force is also added from the patient's right elbow up toward the shoulder.
7. By further adding pressure on the shoulder, side-bending is introduced down to the monitoring finger.
8. Rotation toward the physician's monitoring finger is introduced. If tissue tension increases, rotation away may be attempted.
9. The position is held for 3 seconds and then released, and the dysfunction is reevaluated.

Lateral Rib Dysfunctions or Muscle Hypertonicity



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Counterstrain

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Indications:

1. Acute, subacute, and chronic somatic dysfunctions of articular and/or myofascial origin
2. Adjunctive treatment of systemic complaints with associated somatic dysfunction (e.g., viscerosomatic reflex causing rib dysfunction)

Contraindications:

Absolute:

1. Traumatized (sprained or strained) tissues, which would be negatively affected by the positioning of the patient
2. Severe illness in which strict positional restrictions preclude treatment
3. Instability of the area being positioned that has the potential to produce unwanted neurologic or vascular side effects
4. Vascular or neurologic syndromes, such as basilar insufficiency or neuroforaminal compromise whereby the position of treatment has the potential to exacerbate the condition
5. Severe degenerative spondylosis with local fusion and no motion at the level where treatment positioning would normally take place

Relative:

1. Patients who cannot voluntarily relax, so that proper positioning is difficult
2. Stoic patients who cannot discern the level of pain or its change secondary to positioning
3. Patients who cannot understand the instructions and questions of the physician (e.g., patient 6 months of age)
4. Patients with connective tissue disease, arthritis, Parkinson disease, and so on, in whom positioning for tender point pain reduction exacerbates the distal connective tissue or arthritic problem or no motion is available for positioning

Posterior Thoracic Counterstrain
Spinous Process/Inferolateral Points

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Spinous Process Points



Inferolateral Points (Bilateral)

Posterior Thoracic Counterstrain Spinous Process/Inferolateral Points



Fig 9.70

Treatment Position: E Sa Ra

PT1–PT4 (e.g., T2 tender point right inferolateral aspect T2 spinous process, deviated to the right).

1. The patient lies supine with the neck and upper back off the edge of the table.
2. The physician sits at the head of the table supporting the patient's head to prevent hyperextension of the occipitoatlantal and cervical region.
3. The physician extends to the level of the dysfunction and fine-tunes with side bending and rotation away until the tenderness has been *completely alleviated* or reduced as close to 100% as possible, but at least 70% (Fig. 9.70).

Posterior Thoracic Counterstrain Spinous Process/Inferolateral Points



Fig 9.71

Treatment Position: E Sa Ra

PT5–PT9 (e.g., T6 tender point right inferolateral aspect T6 spinous process, deviated to the right)

1. The patient lies prone, and the physician stands or sits at the side of the tender point.
2. The patient rotates the head/neck to the left (side opposite tender point).
3. The patient's right arm/shoulder (side of tender point) is flexed, resting comfortably on the side of the head and the left arm along the side of the body.
4. The patient's left shoulder is pulled posterior and caudad, which produces extension, rotation away, and side bending away from the side of the tender point and deviated spinous process (Fig. 9.71).
5. The physician fine-tunes through small arcs of motion (more or less extension, rotation, and side bending) until the tenderness has been *completely alleviated* or reduced as close to 100% as possible, but at least 70%.

Posterior Thoracic Counterstrain Transverse Process (Lateral) Points

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Posterior Thoracic Counterstrain Transverse Process (Lateral) Points

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Treatment Position: E Sa RT

PT4-9 Lateral Tender Points

1. The patient lies prone, the head rotated to the side of the tender point, and the physician sits at the head of the table. The physician's forearm is placed under the patient's axilla on the side of the tender point with the hand on the posterolateral chest wall. The physician's forearm lifts patient's shoulder to produce extension and rotation to the side of the tender point and side bends the torso by adding more shoulder abduction (Fig. 9.73).
2. The patient lies prone, the head rotated to the side of the tender point, and physician stands at the side opposite the tender point. The patient's torso may be side bent away and the arm on the side of the tender point abducted to produce even more side bending away. The patient's left shoulder is pulled posterior and cephalad, which produces extension and rotation toward and side bending away from the side of the tender point (Fig. 9.74).
3. The side bending component may vary depending on which specific myofascial structures are involved, the direction of their fibers, or whether there is an articular component to the dysfunction.
4. The physician fine-tunes through small arcs of motion (more or less extension, rotation, and side bending) until the tenderness has been *completely alleviated* or reduced as close to 100% as possible, but at least 70%.

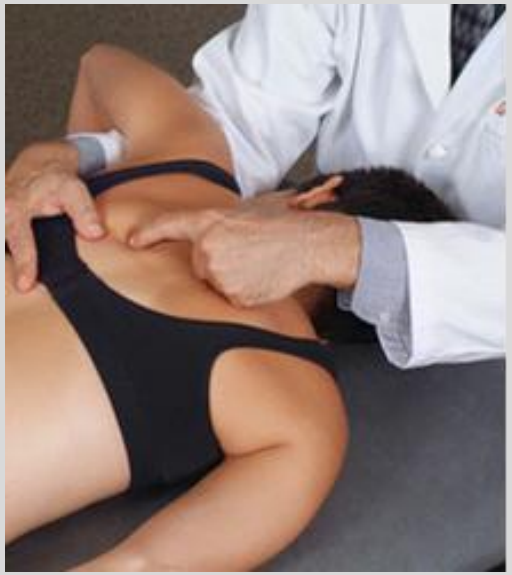


Fig 9.73

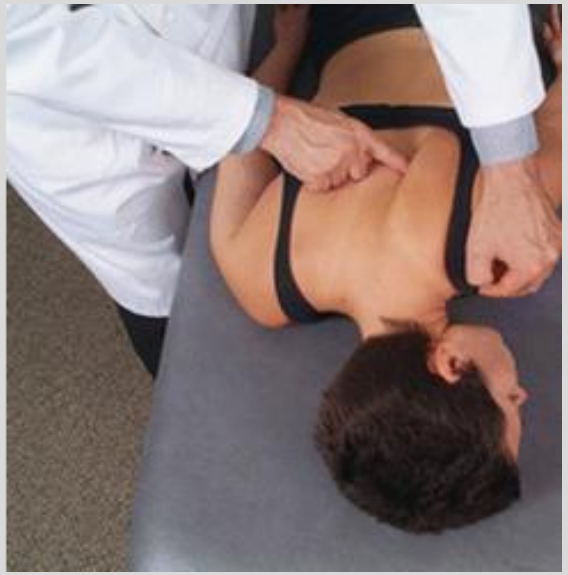


Fig 9.74

Posterior Thoracic Counterstrain Inferolateral and Transverse Process Points PT10-12

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Posterior Thoracic Counterstrain Inferolateral and Transverse Process Points PT10-12

PT 10-12 Spinous Process (Inferolateral): On the inferolateral aspect of the deviated spinous process of the dysfunctional segment.

Treatment Position: e-E Sa Rt (pelvis) Ra (torso)

T10–T12 (e.g., T11 tender point left; T11 **spinous process**, deviated to the left, rotated right).

1. The patient lies prone, and the physician may stand on either side of the tender point. The patient's legs are positioned to the side which produces the greatest reduction of tenderness. The side bending component may vary depending on which myofascial structures are involved, the direction of their fibers, or whether there is an articular component to the dysfunction. The physician grasps the anterior superior iliac spine (ASIS) on the side of the tender point; leans back, gently lifting upward to induce extension and rotation of the pelvis (lower segment) toward the side of the tender point; and rotates the torso (upper segment) away (Fig. 9.76).



Fig 9.76

Posterior Thoracic Counterstrain Inferolateral and Transverse Process Points PT10-12

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PT 10-12 Transverse Process: Posterolateral aspect of the transverse process of the dysfunctional segment.

Treatment Position: e-E Sa Rt (torso) Ra (pelvis)

T10–T12 (e.g., T11 tender point right; T11 **transverse process**, rotated to the right).

1. The patient lies prone, and the physician may stand on either side of the tender point. The patient's legs are positioned to the side which produces the greatest reduction of tenderness. The side bending component may vary depending on which myofascial structures are involved, the direction of their fibers, or whether there is an articular component to the dysfunction. The physician grasps the ASIS on the side opposite the tender point; leans back, gently lifting upward to induce extension and rotation of the pelvis (lower segment) away from the side of the tender point; and rotates the torso (upper segment) toward the side of the tender point (Fig. 9.77).

2. The physician fine-tunes through small arcs of motion (extension, rotation, and side bending) until the tenderness has been completely alleviated or reduced as close to 100% as possible, but at least 70%.



Fig 9.77

Review Material

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***The review material for this lab includes:

Fascial diagnosis to regions MFR thoracic inlet

MFR respiratory diaphragm MFR pelvic diaphragm

Rib raising - supine

Diagnosis and Treatment - OA, cervical, thoracic, ribs

FPR/Still Cervical/OA

Selected lymphatic review techniques from this list are reviewed in full in the OMM handout, but you are responsible for ALL of the techniques listed above for your exams.



Supplemental Material

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Cases

After practicing these new techniques, please complete the CHF cases. You and your lab partner are to complete a different case. There is some helpful supplemental material following each case. You are to:

1. Evaluate your patient and document each of the parts in red text.
2. Diagnose somatic dysfunctions, have a table trainer check your diagnoses and then treat them using OMT. Please call a table trainer over to help you treat your patient if you need help.
3. After both of you have completed the OMT part on each other, work on completing the SOAP notes.
4. Don’t forget patient education.



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